



PROJECT REPORT

A digital companion built for the Indian farmer
— crop diagnosis, fair prices, and insurance
support, in one app.

Submitted to: **The Head, SDPS International
School, Indore**

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Live product: bhoomix.vercel.app

1. Why This Project Was Built

This project did not start as a school assignment looking for a topic. It started from something I have seen with my own eyes.

I come from a farming background — my family belongs to a village, and I have grown up around the everyday reality of farming life. Over time, I noticed the same pattern happening again and again: farmers in my own community were working just as hard every season, but were not getting a fair result for that work. The problem was rarely the crop itself. It was almost always something around the crop — not knowing the right market to sell in, not catching a plant disease early enough, or missing a paperwork deadline for a government scheme they were entitled to.

A second, very local observation confirmed that this gap was real and not just a feeling. I live in Indore. Quick-delivery grocery apps like Blinkit are common here, but they do not deliver genuinely fresh vegetables and fruit — they deliver packaged and frozen food. There is a real, visible gap between the farmer who grows fresh produce a short distance away, and the city household that cannot get anything but frozen substitutes delivered to their door. That gap is exactly the argument for building something that connects farmers directly to buyers, using farmers themselves as the source of supply instead of a frozen-food warehouse.

Those two observations together — farmers losing money for reasons that have nothing to do with how good a farmer they are, and cities that cannot access the fresh produce sitting just outside them — are the reason BhoomiX exists. It is not a hypothetical business case. It is a problem I have watched happen, and a gap I have personally tried to buy fresh food and failed to find.

2. The Problem, Stated Precisely

A farmer typically loses money in three specific, avoidable ways:

Way money is lost	Why it happens
Selling at the wrong market	The same crop can sell for very different prices at nearby markets on the very same day, and a farmer usually has no easy way to compare rates before deciding where to sell.
Catching crop disease too late	By the time a disease is visible to the naked eye without help, meaningful crop loss has often already happened. Expert diagnosis is often a long trip away.
Missing the insurance reporting window	Under India's national crop insurance scheme, a localised loss must be reported within 72 hours. A claim reported late can be rejected regardless of how real the loss was.

Existing crop-focused apps in the market generally solve only the first of these — identifying a plant problem. They do not help with market prices or insurance deadlines, and they are built for the whole country rather than for the reality of any one district.

3. What Was Actually Built

BhoomiX is a working web application, live today at **bhoomix.vercel.app** — not a design mock-up or a slide deck. The features below are all functioning in the deployed product.

Crop Intelligence **LIVE**

A farmer photographs a leaf. The system identifies the crop and its growth stage, names the likely disease, gives a confidence score, and suggests a treatment.

Mandi (Market) Prices **LIVE**

Daily crop prices pulled from the Government of India's open data platform, filterable by crop and by state, so a farmer can compare markets before selling.

Crop Damage / Insurance Reports **LIVE**

Photos are stamped with GPS location and exact time, paired with recorded rainfall, inside a visible 72-hour countdown that matches the real insurance-scheme deadline.

Government Schemes **LIVE**

Ten central and state schemes explained in plain language — who can apply, what the benefit actually is, and a direct link to apply.

Crop Advisory Assistant **LIVE**

An assistant that answers farming questions by text or photo, in the farmer's own language, at any time of day.

Agri Market **LIVE**

Seeds, fertiliser and tools, linked directly from whatever the diagnosis just recommended, so advice and the ability to act on it are in the same place.

AgriNova Mart **LIVE**

Everyday groceries sourced with farmers as the supply, addressing the exact fresh-food gap that quick-delivery apps in Indore currently leave unfilled.

22 Indian Languages + English **LIVE**

The full interface, not just a translated label, works in all 22 languages of the Eighth Schedule of the Constitution, plus English — 23 languages in total.

The One Feature No Competitor Has

Every crop-scanning app in India collects photographs of unhealthy plants. None of them ask what actually cured the plant. Seven days after a diagnosis, BhoomiX asks the farmer one simple question: did the treatment work? That answer is recorded, and it makes the advice given to the next farmer measurably better. This is a dataset that grows more valuable every season and cannot be copied simply by building a similar-looking app.

4. How It Was Built

BhoomiX is a real software product, built using the same category of tools professional companies use, not a static prototype:

- **Frontend:** A responsive web application, so it works on any phone with a browser — no separate app store install required for a farmer to start using it.
- **AI diagnosis:** Crop photographs are analysed by a multimodal AI model that returns a disease name, a confidence score, and a treatment suggestion.
- **Backend and data:** A secure cloud database handles accounts, orders, and diagnosis history, with row-level access rules so one user's data is never visible to another.
- **Live government data:** Mandi prices are fetched directly from India's official open-data platform rather than being manually entered or estimated.
- **Deployment:** The application is deployed to a live public URL, so it can be opened and tested by anyone at any time — including while presenting to judges or reviewers.

Development used AI-assisted, prompt-based coding as an engineering method: describing the required behaviour precisely and iterating quickly, in the same way a professional team uses modern development tools to move faster without lowering the bar on correctness. Every feature listed in Section 3 was individually tested against the live, deployed application before being called complete.

5. Current Status — An Honest Account

This report intentionally avoids overstating where the project is. As of today:

- The product is **fully built and live**, and every feature above can be demonstrated on request, live, not from screenshots.
- The project is at an **early stage of real-world adoption**. It has not yet been piloted with a formal group of farmers, and no revenue has been generated yet.
- Formal user interviews with farmers have **not yet been conducted**. This is a known gap, not a hidden one, and closing it is the immediate next step (see Section 7).
- A distributor tour to identify and vet high-quality seed suppliers, mentioned as a planned activity, has **not yet taken place**.

Being specific about what is proven versus what is planned is a deliberate choice. A report that claims more than what exists would not survive a serious follow-up question, and that is a worse outcome than being upfront now.

6. Why This Matters — Beyond the App

BhoomiX connects to United Nations Sustainable Development Goals through features that already exist, not through aspiration:

SDG 1 — No Poverty: Closing the market-price gap protects income a farmer has already earned.

SDG 2 — Zero Hunger: Early disease detection protects yield before a field is lost.

SDG 8 — Decent Work: Fair price discovery for farmers, and livelihoods for local delivery partners.

SDG 9 — Innovation: Puts AI diagnostics and government open data in a smallholder's pocket.

SDG 10 — Reduced Inequalities: 23-language support removes the literacy barrier that locks many farmers out of digital tools.

SDG 13 — Climate Action: Rainfall is recorded with every damage report, building a genuine climate-risk record over time.

7. Relevance Beyond Agriculture — Pharmacy Departments

BhoomiX is built for farmers, but two parts of the product already generate data that a college Pharmacy department — specifically the subjects of Pharmacognosy and Phytochemistry, which study medicinal and economically important plants — could genuinely use:

1. A live catalogue of medicinal plants, with real sourcing data

The Plant Buyer feature already lists medicinal and economically valuable plants for sale through verified local nurseries, including **Tulsi**, **Ashwagandha**, **Aloe Vera**, and **Stevia** — all standard subjects in a Pharmacognosy syllabus — alongside their real price, unit, and the specific nursery and distance supplying them. This is regional cultivation and availability data that a pharmacy student would otherwise have to collect by hand.

2. A record of organic and biological crop treatments, with outcomes attached

The Organic Treatment section already recommends Neem oil, Trichoderma, and Panchagavya as biological alternatives to chemical pesticides. Combined with the seven-day outcome check described in Section 3, this becomes real efficacy data over time — which of these plant-derived treatments actually worked, on which crop, in which season. That is exactly the kind of applied dataset a pharmacy research project on plant-based crop protection would need, and normally cannot get.

This is not a built feature today — it is an honest observation that the underlying data already exists inside BhoomiX for a different purpose. A future collaboration with a Pharmacy department's Pharmacognosy faculty, treating BhoomiX as a live data source for medicinal-plant sourcing and organic-treatment efficacy, is a realistic and low-cost next step, not a speculative one.

8. Technical Appendix — Stack, Codebase and Security

8.1 — Technology Stack

Layer	What is used
Frontend framework	React 18 with TypeScript, built using Vite
Styling	Tailwind CSS, with a shared component library (shadcn/ui) for consistent controls
Routing	React Router, with role-aware protected routes
State management	React Context, scoped per concern (authentication, cart, language, theme) rather than one global store
Backend database	A managed PostgreSQL database (Supabase) with authentication, serverless functions, and file storage built in
AI — crop & disease diagnosis	Google Gemini, a multimodal model capable of reading an uploaded photograph and returning structured analysis
AI — advisory chat	A vision-capable language model served through NVIDIA's inference platform
Live market data	India's official Government open data platform (data.gov.in), proxied through a serverless function with hourly caching
Weather data	Open-Meteo, a free public weather API requiring no key, chosen after an earlier paid provider's key stopped working
Transactional email	Resend, for password-reset and account email, with delivery events logged back for troubleshooting
Hosting / deployment	Vercel (frontend), auto-deployed on every code change; source controlled with Git

8.2 — Security Measures Actually Implemented

- **Row Level Security on every table.** The database denies access by default; a user can only read or write rows a specific policy explicitly allows them to.
- **No secret ever reaches the browser.** AI keys, the market-data key, and the database's administrative key exist only inside serverless functions the browser calls — never in code the browser downloads and can inspect.
- **Role checks that cannot be bypassed from the client.** Whether an account can create a manager or a delivery partner is enforced inside the server function itself, not by hiding a button in the interface.

- **Delivery-partner privacy by design.** A partner cannot see a customer's phone number or address until they accept that specific order — enforced by the database query itself returning blank fields, not by a condition in the app that a modified request could skip.
- **Anti-enumeration on password reset.** The "forgot password" form gives an identical response whether or not an account exists for that email, so it cannot be used to discover who has an account.
- **Signed, replay-protected webhooks.** Email delivery events are cryptographically verified (HMAC-SHA256) before being trusted, and any event older than five minutes is rejected outright, so a captured request cannot be replayed later.

8.3 — An Honest Gap: Automated Testing

There is currently no automated test suite (no unit or integration tests) in the codebase. Every feature described in this report was verified manually against the live, deployed application, which is reliable for confirming a feature works today but does not automatically catch a regression introduced by a future change. Adding automated tests for the highest-risk flows — sign-in, payment-adjacent cart logic, and the insurance-report deadline — is a known next step, listed here rather than left unmentioned.

9. What Happens Next

9.1 — A Field Research Tour

The next planned phase is a short, structured tour to two or three nearby farming districts, with two goals: (1) speak directly with farmers to validate that the product solves their real problem, and (2) identify and evaluate genuine high-quality seed distributors who could be recommended inside the Agri Market feature, so that a farmer buying seed through BhoomiX is buying something verified, not a random listing.

9.2 — Government Schemes and Subsidy Support

BhoomiX already surfaces existing government agricultural schemes to farmers. The next step is to research whether the project itself, as a genuine agri-tech initiative, qualifies for any state or central government startup, MSME, or agri-innovation support programme, and to formally apply where eligible.

9.3 — A Simple, Honest Business Model

The app is free for the farmer, on purpose — a farmer unsure whether advice is affordable will not ask for it. Revenue is planned to come from the commerce the advice creates: a small commission on marketplace orders, paid listings for local seed and input dealers, and delivery margin on fulfilled orders. This is a model that only earns once it is genuinely useful to a farmer, not before.

9.4 — A Long-Term Data Advantage

Over three to five years, the outcome-tracking loop described in Section 3 builds something no competitor can quickly copy: a real record, season after season, of which treatments actually worked for which crops in which conditions. In the long run, this kind of aggregated, anonymised agricultural outcome data is valuable to banks and lenders assessing agricultural loan risk more accurately than they can today — a bank could, with farmer consent and proper privacy safeguards, use verified outcome history to offer better loan terms to farmers with a demonstrated track record. This is a future direction, not a current feature, and would require careful work on consent and data protection before any such use.

10. Closing Statement

This project was not built to win a competition first and find a reason for it afterward. It was built because a real gap was visible from where I stand — in a farming family, in a city that cannot get fresh food delivered. Everything described in this report can be opened and tested live, right now, at bhoomix.vercel.app. That is the standard this report has tried to hold itself to throughout: say only what is true, and be clear about what is still ahead.

